A model of univalence over a base category with an interval

posets and simplicial sets as instances

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Abstract

We present a model of dependent type theory with a univalent universe of fibrant types, over an abstract base category with an interval object and a presheaf of cofibrations whose elements are decidable sieves. Fibrancy is a structure: a uniform composition operation along lines. All the structures of the model are instances of one notion, a relative contractibility structure, and they are transported along maps of contexts by one lemma. Following Voevodsky, every construction on a family of data (the glued type of Theorem 6.2, its classifier, the univalence operation) is made once on a universal family and obtained by substitution along a classifying map; naturality is then the composition law of substitution together with one equation of data. The two instances are the category of finite non-empty posets, where the argument is constructive, and the simplex category Δ, where excluded middle is used at exactly one point, the passage from horn filling to box filling. Finally, we give identity types with a strict computation rule: Swan's, for both instances, equivalent to the path type; and, on Δ, the path type itself, using excluded middle a second time.

Contents

1 Metatheory **2**

2 The base **3**
2.1 The category and the interval . . . 3
2.2 Presheaves, types, lines . . . 3
2.3 Cofibrations . . . 3

3 Relative contractibility **4**
3.1 Partial elements . . . 4
3.2 The notion . . . 4
3.3 Instances . . . 4
3.4 Moving a structure along a map . . . 5

4 Fibrations **6**

5 Small types and the universe **7**
5.1 Small types over large contexts . . . 7
5.2 The universe . . . 7

6 Gluing: Theorem 7 **8**
6.1 Ext(α) 8
6.2 The universal glue data 9

7 Univalence and the fibrancy of the universe **10**
7.1 The univalence operation 10
7.2 The universe is fibrant 12

8 Instances **12**
8.1 Posets 12
8.2 Simplicial sets 13

9 Identity types **14**
9.1 Swan's identity type 14
9.2 The path type on Δ 15

A Correspondence with the formalisation **17**

1 Metatheory

We work in dependent type theory with Π, Σ, identity types, natural numbers, Booleans, and a cumulative hierarchy of universes. We assume function extensionality and univalence for the universes of the metatheory; both may be used anywhere. (Function extensionality follows from univalence; we use it directly, and the development is also obtained from univalence alone by one instance, Appendix.) Logic is constructive, with one exception: in Section 8.2, and only there, we use excluded middle for propositions,

$$\mathrm{lem}: \ \Pi(P : \mathrm{Prop})\ P + \neg P.$$

The axiom of choice is never used (Remark 8.4).

Iterative sets. For each universe level a we use the type V_a of iterative sets (Aczel's sets as well-founded trees, with extensional equality; in univalent type theory, the type of well-founded trees whose branching maps are injective). An element $v : \mathsf{V}_a$ has a type of elements $\mathrm{Ix}(v)$, which is a set. Every set in the model is *coded*: the presheaves of the model are families of iterative sets, and two types of the model are equal exactly when their codes are equal in V.

Remark 1.1 (strictness). Equality in V_a is extensional, as in set theory, not univalent: two isomorphic codes are in general different elements. The metatheory is univalent, but this does not make isomorphic types of the model equal. All uniformity conditions of the model (the uniformity of a composition, the naturality of the univalence operation, the stability of Ext under substitution) are *equalities of codes*. Never an isomorphism, never a relation.

Each level a gives a level a^+ with V_a coded in V_{a^+}: a code $v : \mathsf{V}_a$ has a code $\mathrm{up}(v) : \mathsf{V}_{a^+}$ with $\mathrm{Ix}(\mathrm{up}(v))$ and $\mathrm{Ix}(v)$ the same set. Types of the model live at level a; the universe and the contexts built from it live at level a^+.

2 The base

2.1 The category and the interval

The base is a small category $\mathcal{C}$ with an object $[1]$ and, for every object X, two maps $e_0, e_1 : X \to [1]$, natural in X (the constant maps at the end points). We write $\mathbb{I} = y[1]$ for the presheaf represented by $[1]$, so $\mathbb{I}(W) = \mathrm{Hom}(W, [1])$. We do not assume that $\mathcal{C}$ has products: posets and Δ are both instances, and Δ has no product $X \times [1]$.

Connections. There are two maps $c_0, c_1 : \mathbb{I} \times \mathbb{I} \to \mathbb{I}$ (on a stage W: for $i, j : W \to [1]$, a map $c_k(i, j) : W \to [1]$), natural in W, with

$$c_k(e_k, j) = e_k, \qquad c_k(i, e_k) = e_k, \qquad c_k(e_{1-k}, j) = j, \qquad c_k(i, e_{1-k}) = i.$$

In both instances, for the order $e_0 < e_1$, $c_0 = \min$ and $c_1 = \max$, computed pointwise on maps $W \to [1]$.

2.2 Presheaves, types, lines

A *context* is a presheaf Γ on $\mathcal{C}$ with values in coded sets: a family of codes $\Gamma(X)$ with restrictions $\rho \mapsto \rho f$. A *type* A over Γ is a family of codes $A(X, \rho)$ for $\rho \in \Gamma(X)$ with restrictions $a \mapsto af : A(X, \rho) \to A(W, \rho f)$, functorial. We write E_A (or $\Gamma.A$) for the total presheaf of the pairs (ρ, a). A substitution $s : \Delta \to \Gamma$ is a natural transformation, and $A[s](X, \rho) = A(X, s\rho)$.

Definition 2.1 (lines)**.** A *line* of Γ at X is a family $\gamma(W, f, j) \in \Gamma(W)$ for $f : W \to X$ and $j \in \mathbb{I}(W)$, natural: $\gamma(W, f, j) \cdot g = \gamma(V, fg, jg)$. Equivalently a map $yX \times \mathbb{I} \to \Gamma$. The lines form a presheaf $\Gamma^{\mathbb{I}}$, with evaluations $\gamma \mapsto \gamma(e) = \gamma(X, \mathrm{id}, e)$ for $e = e_0, e_1$ and constant lines $\rho \mapsto (W, f, j) \mapsto \rho f$. A line in A over a line γ of Γ is a line of E_A lying over γ; these form a type $A^{\mathbb{I}}$ over $\Gamma^{\mathbb{I}}$.

This is the path object of a presheaf, defined without products in $\mathcal{C}$ (the exponential law holds by definition). On posets $yX \times \mathbb{I} = y(X \times [1])$, so a line at X is an element of $\Gamma(X \times [1])$.

2.3 Cofibrations

Definition 2.2 (cofibrations)**.** For each object X we are given a set $\Phi(X)$ with a map sending $\varphi \in \Phi(X)$ to a *decidable sieve* on X: a decidable property "f lands in φ" of maps $f : W \to X$, closed under precomposition. Two elements with the same maps landing in them are equal. Φ is a presheaf: $f^*\varphi$ is the sieve of the g with fg in φ. We assume:

1. the empty sieve $\bot$ and the total sieve $\top$ are in Φ; whether $\varphi = \top$ is decidable (it is whether id_X lands in φ);
2. $\Phi(X)$ is closed under finite unions;
3. the points: the sieves $(j = e_0)$ and $(j = e_1)$ on W, for $j : W \to [1]$, are in $\Phi(W)$;
4. $\forall$ along the interval: for a sieve L of $yX \times \mathbb{I}$ (a decidable family $L(W, f, j)$, closed under restriction) given by elements of Φ on the generalised points, there is $\forall L \in \Phi(X)$ with

$$f \in \forall L \iff L(V, fg, j) \text{ for all } g : V \to W,\ j \in \mathbb{I}(V).$$

Since an element of Φ *is* a sieve, a cofibration given by several faces is one element, their union; a list of faces is only a notation for it.

Remark 2.3. On posets the subsets of X are not closed under union as sieves: on [1], the union of the subsets $\{0\}$ and $\{1\}$ is the whole of [1] as a subset, but the identity of [1] lands in neither. So the sieves of the poset instance are the decidable sieves of maps, as on Δ.

A *cofibration* over Γ is a natural family $\psi_\rho \in \Phi(X)$ for $\rho \in \Gamma(X)$. The point ρ is *full* if $\psi_\rho = \top$; this is decidable, and f is full (ρf is full) exactly when f lands in ψ_ρ. We write $\Gamma|\psi$ for the context of full points.

3 Relative contractibility

3.1 Partial elements

Definition 3.1. Let S be a type over Γ and $\rho \in \Gamma(X)$. A *partial element* of S at ρ is a pair (φ, u) with $\varphi \in \Phi(X)$ and u a family $u_f \in S(W, \rho f)$ for $f : W \to X$ landing in φ, natural: $u_f \cdot g = u_{fg}$. Its *restriction* along $h : Y \to X$ is $(h^*\varphi,\ g \mapsto u_{hg})$ at ρh.

Partial elements at ρ form a presheaf Partial(S) over Γ (cours6: $\tilde{A}$). Two partial elements are equal when their sieves are equal and their values agree; the values do not depend on the proof that f lands.

3.2 The notion

Definition 3.2 (relative contractibility structure)**.** Let S and T be types over Γ and $m : S \to T$ a map over Γ. A *relative contractibility structure* $k \in \mathrm{RC}(\Gamma, S, T, m)$ assigns to each $\rho \in \Gamma(X)$ and each partial element (φ, u) of S at ρ an element $k(\rho, \varphi, u) \in T(X, \rho)$ such that

1. (sides) $k(\rho, \varphi, u) \cdot f = m(u_f)$ for f landing in φ;
2. (uniformity) $k(\rho, \varphi, u) \cdot h = k(\rho h, h^*(\varphi, u))$ for $h : Y \to X$.

The two conditions are propositions, so:

Lemma 3.3 (extensionality)**.** *Two relative contractibility structures with the same values are equal.*

Remark 3.4 (property gives structure). A structure k is a map Partial$(S) \to T$ over Γ restricting to m on the sieve; uniformity is its naturality. It is one map out of the generic partial element. No choice is involved.

3.3 Instances

Definition 3.5.

1. A *contractibility structure* on S is $\mathrm{Contr}(\Gamma, S) = \mathrm{RC}(\Gamma, S, S, \mathrm{id})$: every partial element extends to a total one, uniformly.
2. For $\alpha : B \to A$ over Γ, the *strict fibre* of α at $(\rho, a) \in E_A$ is the set of $b \in B(\rho)$ with $\alpha b = a$; this is a type SF_α over E_A. A *trivial fibration structure* on α is $\mathrm{Contr}(E_A, \mathrm{SF}_\alpha)$.
3. *Filling at e*: $\mathrm{Fill}_e(A)$ is a trivial fibration structure on the evaluation $\mathrm{ev}_e : A^{\mathbb{I}} \to A[\mathrm{ev}_e]$ over $\Gamma^{\mathbb{I}}$: a partial line with a total end at e extends to a total line with that end.
4. *Composition from e to e'*: let L_e be the type, over the context $\Gamma^{\mathbb{I}}.A[\mathrm{ev}_e]$ of pairs (line γ, element a over $\gamma(e)$), of the lines in A over γ ending at a at e, and $m = \mathrm{ev}_{e'} : L_e \to A[\mathrm{ev}_{e'}]$. Then
$$\mathrm{Comp}_{e \to e'}(A) = \mathrm{RC}\big(\Gamma^{\mathbb{I}}.A[\mathrm{ev}_e],\ L_e,\ A[\mathrm{ev}_{e'}],\ \mathrm{ev}_{e'}\big).$$

A composition takes a partial line with a total start a and gives an element at the other end that agrees with the ends of the sides.

5. For $\alpha : B \to A$ over Γ, the *homotopy fibre* HFib_α over E_A has at (ρ, a) the pairs (b, ℓ) with $b \in B(\rho)$ and ℓ a line in A over the constant line at ρ with $\ell(e_0) = \alpha b$ and $\ell(e_1) = a$. An *equivalence structure* on α is $\mathrm{Equiv}(\alpha) = \mathrm{Contr}(E_A, \mathrm{HFib}_\alpha)$.

Remark 3.6. Because a partial element is given on one sieve, a structure cannot distinguish two presentations of the same sieve. No condition of invariance under re-presentation (inserting an empty face, permuting faces) is needed.

3.4 Moving a structure along a map

Construction 3.7 (move)**.** Let $s : \Delta \to \Gamma$ be a map of contexts, possibly from a context of level a to one of level a^+. Let $m' : S' \to T'$ over Δ and $m : S \to T$ over Γ, and let $\sigma : S' \to S[s]$ and $\tau : T' \to T[s]$ be maps over Δ with $\tau m' = m\sigma$, τ a bijection on each fibre. For $k \in \mathrm{RC}(\Gamma, S, T, m)$ define

$$s^*k(\rho, \varphi, u) = \tau^{-1}\big(k(s\rho, \varphi, \sigma u)\big).$$

Lemma 3.8. $s^*k \in \mathrm{RC}(\Delta, S', T', m')$.

Proof. On a side f: $s^*k(\ldots) \cdot f = \tau^{-1}(k(s\rho, \varphi, \sigma u) \cdot f) = \tau^{-1}(m\sigma u_f) = \tau^{-1}\tau m' u_f = m' u_f$, using that τ and τ^{-1} commute with restriction. Uniformity follows in the same way from the naturality of s, σ and τ. □

Lemma 3.9 (laws)**.** *Moving along the identity with identity maps is the identity. Moving along s and then along t is moving along st with the composite maps. Moving commutes with substitution, and is a congruence in its data.*

Proof. In each case the two structures have the same values, by the definition of s^*k; conclude by Lemma 3.3. □

Lemma 3.10 (transport is a move)**.** *Let $P : A = A'$ be an equality of types over Δ and c a fibration structure on A. Then $(A, c) = (A', c')$, where c' is c moved along the cartesian map whose fibre maps are transport along P. These fibre maps may be taken to be transport along any equality of the fibres, since V is a set.*

Proof. By induction on P; at reflexivity the cartesian map is the identity, and Lemma 3.9 gives $c' = c$. □

In practice the move is used at the level of types. A *cartesian map* $A' \to A$ over $s : \Delta \to \Gamma$ is a map $A' \to A[s]$ over Δ which is a bijection on each fibre. It induces the maps σ and τ for every structure of the model built from A (for compositions: on partial lines and on end points), so that a cartesian map moves Comp and Fib in one step. All the transfers of the model are instances of a cartesian map: substitution along s (the identity $A[s] \to A[s]$); a type of level a read at level a^+ (the fibres are the same sets); and a type isomorphic fibrewise to the substitution of another.

4 Fibrations

Definition 4.1. A *fibration structure* on A over Γ is a pair of compositions

$$\mathrm{Fib}(\Gamma, A) = \mathrm{Comp}_{e_0 \to e_1}(A) \times \mathrm{Comp}_{e_1 \to e_0}(A).$$

A *fibrant type* over Γ is a pair (A, c) with $c \in \mathrm{Fib}(\Gamma, A)$.

The composition is the stored structure; filling is derived.

Proposition 4.2 (filling from composition)**.** *A composition* $\mathrm{Comp}_{e \to e'}(A)$ *gives* $\mathrm{Fill}_e(A)$*, uniformly. The construction commutes with moves: the filling of a moved composition is the moved filling (Construction 3.7).*

Proof. Given a partial line (φ, u) in A over γ with total end a at e, consider at the stage (W, f, i) the line $j \mapsto \gamma(f, c(i, j))$ along the connection that is constant at e. Its partial line has the sides of u connected, and a side on $(i = e)$ given by a. Its composite at e' is the value of the filler at i. The connection laws give the end at e and the sides; uniformity is that of the composition. For moves: the filling problem at a point is formed from the data at that point only, so the problem of the moved composition is the moved problem; conclude by Lemma 3.3. □

Lemma 4.3 (end of the filler)**.** *At $i = e'$ the filler of Proposition 4.2 is the composite of the composition it is built from.*

Proof. At $i = e'$ the extra side $(i = e)$ is empty, since e and e' are disjoint, so the sieve of the problem is the original one; by the connection laws the line $j \mapsto \gamma(f, c(e', j))$ is γf itself. The two problems are equal, hence so are their composites. □

Proposition 4.4 (substitution)**.** *For $s : \Delta \to \Gamma$, a fibration structure on A moves to one on $A[s]$ (Construction 3.7, with lines moved along $s^{\mathbb{I}}$). The laws $c[\mathrm{id}] = c$ and $c[st] = c[s][t]$ hold strictly (Lemma 3.9).*

Theorem 4.5 (type formers)**.** *Fibrant types are closed under Σ, Π and path types, naturally in Γ (for $s : \Delta \to \Gamma$, the structure on $\Sigma(A[s], B[s])$ is the move of the one on $\Sigma(A, B)$ along the cartesian map $\Sigma(A[s], B[s]) \to \Sigma(A, B)$ over s, and likewise for Π and paths):*

$$\mathrm{Fib}(\Gamma, A) \to \mathrm{Fib}(\Gamma.A, B) \to \mathrm{Fib}(\Gamma, \Sigma_A B), \qquad \mathrm{Fib}(\Gamma, A) \to \mathrm{Fib}(\Gamma.A, B) \to \mathrm{Fib}(\Gamma, \Pi_A B),$$
$$\mathrm{Fib}(\Gamma, A) \to \mathrm{Fib}(\Gamma, \mathrm{Path}_A(a, b)).$$

Proof. Σ: fill the first component in A, then the second in B over the line of $\Gamma.A$ so obtained (a line in A over a line in Γ is a line in $\Gamma.A$). Π: for a line of arguments ending at a given argument, fill it backwards in A, apply the partial line of functions to the filled line, and compose in B. Path: a partial line of paths is a partial line in A over the lines $\gamma(f, j)$ with the two end conditions as further sides on $(i = e_0)$ and $(i = e_1)$; these three sides are joined into one sieve (their union), and one composes in A. As for filling (Proposition 4.2), the problem is formed from the input restricted along the identity, so that one uniformity argument gives both its naturality and its uniformity.

Naturality: each construction is formed pointwise from the structures of A and B and from filling, which commutes with moves (Proposition 4.2); so the structure on the substituted type and the moved one have the same values, and are equal by Lemma 3.3. The form with a cartesian map avoids an equality of types $\Sigma(A[s], B[s]) = \Sigma(A, B)[s]$, which holds only up to an identification of the actions; with it, Lemma 3.10 gives the substitution form. □

Proposition 4.6 (transport is an equivalence)**.** *Let A be fibrant over Γ and γ a line of Γ at X. The end at e' of the filler from e is a map $\mathrm{tr}_\gamma : A(\gamma(e)) \to A(\gamma(e'))$ over yX, natural in γ, with an equivalence structure, natural in γ.*

Proof. The construction is made once over the universal line, over $\Gamma^{\mathbb{I}}$, and substituted along $\gamma : yX \to \Gamma^{\mathbb{I}}$. The equivalence structure extends a partial fibre element (b, ℓ) by filling, along the given line, the paths of the fibre; this is a composition in a path type over a line. □

5 Small types and the universe

5.1 Small types over large contexts

Definition 5.1. Let G be a context of level a^+. A *small type* over G is a type whose fibres are codes of level a: $A(X, x) : \mathsf{V}_a$ for $x \in G(X)$. We write $\mathrm{STy}(G)$; $\mathrm{up}A$ is A read as a type of level a^+ (the same fibres). For a context D of level a and a map $s : D \to G$, the substitution s^*A is the type of level a with $(s^*A)(X, \rho) = A(X, s\rho)$.

Lemma 5.2 (composition law)**.** *For $t : T \to D$, $(s^*A)[t] = (st)^*A$, as an equality of types.*

Proof. The fibres agree by definition; the restrictions agree because they differ only by the proof of naturality used to transport along, and the contexts are sets. □

This is Voevodsky's chosen pullback: substitution of small types is strictly functorial, once for all small types. A structure on $\mathrm{up}A$ (a fibration structure, an equivalence structure) moves to s^*A by Construction 3.7, with identity fibre maps, and its laws are those of Lemma 3.9.

5.2 The universe

Definition 5.3 (classifier)**.** A *fibrancy classifier* assigns to each context Γ (of level a) and type A over Γ a set $\mathrm{Cl}(\Gamma, A)$, with restrictions $c \mapsto c[s]$ along $s : \Delta \to \Gamma$ satisfying

$$(A, c)[\mathrm{id}] = (A, c), \qquad (A, c)[st] = (A, c)[s][t]$$

as equalities in $\Sigma_A \mathrm{Cl}(\Gamma, A)$, and such that $\mathrm{Cl}(yX, A)$ has a code in V_{a^+}.

Definition 5.4 (the universe)**.** $\mathsf{U}(X) = \Sigma(A \in \mathrm{Ty}_a(yX))\, \mathrm{Cl}(yX, A)$, with restriction along $f : Y \to X$ given by substitution along yf. This is a presheaf of codes of level a^+. $\mathsf{El} \in \mathrm{STy}(\mathsf{U})$ is the small type with $\mathsf{El}(X, (A, c)) = A(X, \mathrm{id}_X)$.

A type A over Γ with $c \in \mathrm{Cl}(\Gamma, A)$ has a *code map* $\ulcorner A \urcorner : \Gamma \to \mathsf{U}$, $\rho \mapsto (A[\hat{\rho}], c[\hat{\rho}])$, where $\hat{\rho} : yX \to \Gamma$ is the Yoneda map of ρ; and $\ulcorner A \urcorner^* \mathsf{El} = A$ as types. A small type over a context of codes is read by El: for any $c : \Delta \to \mathsf{U}$, $c^*\mathsf{El}$ is "El along c".

Hypothesis 5.5 (El is fibrant)**.** $\mathrm{up}\mathsf{El}$ has a fibration structure over U.

Then every type of the form $c^*\mathsf{El}$ is fibrant, by moving El's structure along c; the structures so obtained are natural in c by Lemma 3.9. Every fibration structure used by the constructions below is of this form.

Remark 5.6. The classifier is a parameter because the two instances differ: on posets $\mathrm{Cl} = \mathrm{Fib}$ and Hypothesis 5.5 holds because $\mathbb{I}$ is tiny (Section 8.1); on Δ, $\mathbb{I}$ is not tiny, Cl is horn filling, and Hypothesis 5.5 uses excluded middle (Section 8.2).

6 Gluing: Theorem 7

6.1 $\mathrm{Ext}(\alpha)$

Definition 6.1. Let A be a type over Γ, ψ a cofibration over Γ, B a type over $\Gamma|\psi$ and $\alpha : B \to A|\psi$ a map. For $\rho \in \Gamma(X)$:

- if ρ is full, $\mathrm{Ext}(\alpha)(X,\rho) = B(X,\rho)$;
- otherwise $\mathrm{Ext}(\alpha)(X,\rho)$ is the set of pairs (a,b) with $a \in A(X,\rho)$ and b a natural family $b_f \in B(W,\rho f)$ for the full $f : W \to X$, with $\alpha(b_f) = af$.

Restriction is by cases on the source and the target (a full point restricts to a full point). The map $\mathrm{ext} : \mathrm{Ext}(\alpha) \to A$ is α on full points and the first projection otherwise. The map glue sends a pair (a,b) to b_{id} at a full point and to (a,b) otherwise; so $\mathrm{ext}(\mathrm{glue}(a,b)) = a$ in both cases.

On ψ, $\mathrm{Ext}(\alpha)$ *is* B and ext *is* α: as codes, not up to isomorphism.

Theorem 6.2 (Theorem 7)**.** *Let A and B be fibrant and $e_\alpha \in \mathrm{Equiv}(\alpha)$. Then* $\mathrm{Ext}(\alpha)$ *has a fibration structure and* ext *an equivalence structure, naturally in* Γ*. On full points the composition of* $\mathrm{Ext}(\alpha)$ *is that of B and the equivalence structure of* ext *is e_α.*

Proof: ext *is an equivalence.* Let $w = (\rho, a) \in E_A(X)$ and (φ, p) a partial element of $\mathrm{HFib}_{\mathrm{ext}}$ at w.

1. For full $f : W \to X$, p restricted along f is a partial element of HFib_α at wf, since $\mathrm{Ext}(\alpha)$ is B there; let $\theta_f \in \mathrm{HFib}_\alpha(wf)$ be its extension by e_α. Then θ is natural (uniformity of e_α), equals p_f on φ (sides of e_α, with the identity in the restricted sieve), and does not depend on the proof that f is full.
2. Consider the partial line in A over the constant line at ρ, with total end w at e_1, with sides: the lines of p on φ, and the images in A of the lines of θ on ψ_ρ. They agree where both are defined, by (1). Fill it by $\mathrm{Fill}_{e_1}(A)$; call the filler F.
3. Let $a_0 = F(e_0)$ and b_f the B component of θ_f. Then $\alpha(b_f) =$ the line of θ_f at $e_0 = F(f, e_0) = a_0 f$, so (a_0, b) is a pair, and $v_0 = (\mathrm{glue}(a_0, b), F) \in \mathrm{HFib}_{\mathrm{ext}}(w)$.

v_0 restricts to p on φ by (1) and the sides of the filler, and is uniform because every step is. At a full ρ, $\psi_\rho = \top$, so F is the line of θ_{id}, which is the extension by e_α. No $\forall$ is needed: the line of Γ is constant. □

Proof: composition. Let γ be a line of Γ at X, x an element of $\mathrm{Ext}(\alpha)$ over $\gamma(e)$, and (φ, u) a partial line in $\mathrm{Ext}(\alpha)$ over γ ending at x. Let $\delta = \forall_\gamma \psi$: the maps f such that $\gamma(fg, j)$ is full for all (g, j).

1. On δ the problem is a problem of B over a line of $\Gamma|\psi$; fill it in B: a partial line β in B on δ.
2. In A: the partial line with sides $\mathrm{ext} \circ u$ on φ and $\alpha \circ \beta$ on δ, total end $\mathrm{ext}(x)$ at e, is filled; let a_1 be its end at e'.
3. At $\gamma(e')$, on $\psi_{\gamma(e')}$, the ends of u (on φ) and of β (on δ) give a partial element of the homotopy fibre of α at a_1 (with constant paths); extend it by e_α, on the sieve $\psi_{\gamma(e')}$, to (b_1, ω).
4. Fill in A the line ω backwards from a_1, with sides ext of the ends of u on φ, of β on δ (constant lines), and ω on $\psi_{\gamma(e')}$; its end a_2 satisfies $\alpha(b_1) = a_2$ on $\psi_{\gamma(e')}$.
5. The composite is $\mathrm{glue}(a_2, b_1)$.

It agrees with the sides of u by construction, and is uniform because $\forall$, the fillers and e_α are. If γ is full everywhere, $\delta = \top$ and the composite is B's. □

6.2 The universal glue data

Definition 6.3 (glue data)**.** The presheaf GD of level a^+ has as points at stage I the tuples $d = (A, \varphi, B, \alpha, e)$ where

- $A \in \mathsf{U}(I)$ and $\varphi \in \Phi(I)$;
- B assigns to each $f : W \to I$ landing in φ a code $B_f \in \mathsf{U}(W)$, natural;
- α is a map $\mathsf{El}(B) \to \mathsf{El}(A)$ over the part of yI landing in φ, natural;
- e is an equivalence structure on α (for El's fibrations).

Restriction along $h : J \to I$ is precomposition; its laws are equalities of codes.

The restriction of the component e along h is its move (Construction 3.7) along the cartesian map of triples χ_h from $(A, B, \alpha) \cdot h$ to (A, B, α) over yh, whose fibre maps are identities; the laws of restriction are those of Lemma 3.9.

A structure over a presheaf can be given locally, point by point.

Lemma 6.4 (gluing structures along points)**.** *Let $m : S \to T$ over a presheaf D. Suppose that for each $x \in D(Z)$ we are given a context C_x with a map $\iota_x : C_x \to D$, a point $o_x \in C_x(Z)$ with $\iota_x(o_x) = x$, and a structure $k_x \in \mathrm{RC}(C_x, S[\iota_x], T[\iota_x], m[\iota_x])$. Suppose they are coherent: for every $c \in C_x(Y)$ and every partial element u at $\iota_x(c)$, the extension given by k_x at c and the one given by $k_{\iota_x(c)}$ at $o_{\iota_x(c)}$ have the same image in T (in particular for $c = o_x \cdot g$). Then*

$$k(x, \varphi, u) = k_x(o_x, \varphi, u)$$

is a structure $k \in \mathrm{RC}(D, S, T, m)$, the unique one agreeing with every k_x at o_x; moreover k moved along ι_x is k_x.

Proof. The sides of k are those of the k_x. Uniformity along g is the uniformity of k_x from o_x to $o_x \cdot g$ followed by the coherence. Uniqueness holds by Lemma 3.3. □

The case $C_x = yZ$, $o_x = \mathrm{id}_Z$ says that a structure is determined by its values at the generic points. Stating the coherence in T means that no transport between the types $S[\iota_x][g]$ and $S[\iota_{x \cdot g}]$ is needed.

Over GD there is the *universal glue datum*: A_U, ψ_U, B_U, α_U, e_U, where A_U and B_U are El along the components A and B, and $\psi_\mathsf{U}(d) = \varphi_d$. The structure e_U is glued by Lemma 6.4 from the points' own structures: for $d \in \mathrm{GD}(Z)$, C_d is the context of the equivalence problems of α_d, ι_d is the cartesian map $(f, q) \mapsto (d \cdot f, q)$ to the universal one, and $k_d = e_d$; the coherence is the definition of restriction in GD. Theorem 6.2 at level a^+ gives $\mathrm{Ext}_\mathsf{U} = \mathrm{Ext}(\alpha_\mathsf{U})$ with its fibration and the equivalence structure of ext_U, *once*.

Construction 6.5 (the small copy)**.** $\mathrm{SA} \in \mathrm{STy}(\mathrm{GD})$ has at $d \in \mathrm{GD}(I)$ the fibre at the identity of $\mathrm{Ext}(\alpha_d)$ over yI, where α_d is the datum d read over yI. It is fibrewise isomorphic to Ext_U (the extents agree: f is full for d iff it lands in φ_d), so it carries Ext_U's fibration structure, map to A_U and equivalence structure, moved by Construction 3.7.

As in Voevodsky's universe categories, glue data over a context are maps into the presheaf of glue data.

Definition 6.6 (glue data over a context)**.** Glue data over Γ are a map $\ell : \Gamma \to \mathrm{GD}$; substitution along $s : \Delta \to \Gamma$ is composition, $\ell[s] = \ell \circ s$.

Remark 6.7. By Yoneda, such an ℓ is the same as componentwise data over Γ: a code map $A : \Gamma \to \mathsf{U}$, a cofibration ψ, a code map B over the extent, a map α of their El's and an equivalence structure e (at $\rho \in \Gamma(X)$, $\ell\rho$ is the datum of these along ρ, read over yX). This correspondence is not needed below.

Definition 6.8. $\mathrm{Ext}(\ell) = \ell^*\mathrm{SA}$, with the fibration, the map ext to A and the equivalence structure of SA moved along ℓ (Construction 3.7). Theorem 6.2 is thereby proved once, at the universal data, and holds for all glue data by moving.

Lemma 6.9 (comparison)**.** *The fibre of $\ell^*\mathrm{SA}$ at ρ is the fibre of the direct definition of Section 6 for the datum $\ell\rho$: $B_{\ell\rho}(\mathrm{id})$ if the identity is in $\varphi_{\ell\rho}$, and the glued fibre otherwise.*

Proof. The fibre is that of SA at $\ell\rho$, the fibre at the identity of Ext of the datum $\ell\rho$ over yX. Conclude by the cases on fullness. □

Theorem 6.10 (naturality of Theorem 6.2)**.** *For $s : \Delta \to \Gamma$, there are cartesian maps $\mathrm{Ext}(\ell \circ s) \to \mathrm{Ext}(\ell)$ and $A(\ell \circ s) \to A(\ell)$ over s, with identity fibre maps, commuting with* ext*; along them the fibration and the equivalence structure of $\mathrm{Ext}(\ell \circ s)$ are the moves of those of $\mathrm{Ext}(\ell)$. (With Lemma 3.10 this gives the substitution form $\mathrm{Ext}(\ell)[s] = \mathrm{Ext}(\ell \circ s)$, as for the type formers.)*

Proof. By Definition 6.8 the structures of $\mathrm{Ext}(\ell)$ and $\mathrm{Ext}(\ell s)$ are moves of those of SA along ℓ and ℓs; the cartesian maps are those of Lemma 5.2, and the composite and congruence laws of Lemma 3.9 give the equalities. □

7 Univalence and the fibrancy of the universe

7.1 The univalence operation

Definition 7.1. Over U, Ua is the type with, at $F \in \mathsf{U}(X)$, the triples (B, α, e): $B \in \mathsf{U}(X)$, $\alpha : \mathsf{El}(B) \to \mathsf{El}(F)$ over yX, and $e \in \mathrm{Equiv}(\alpha)$. Restriction along f is substitution along yf on B and α; on e it is the move (Construction 3.7) along the cartesian map of triples over yf, as for GD. Its laws are equalities of codes for B and α, and Lemma 3.9 for e. $\mathrm{Ua}(F)$ is $\Sigma(Y : \mathsf{U})\,\mathrm{Equiv}(Y, F)$.

Equivalently, by Yoneda, $\mathrm{Ua}(F)$ at X is the set of glue data $d \in \mathrm{GD}(X)$ with $A_d = F$ and $\varphi_d = \top$, and restriction is that of GD; its laws are then those of GD.

Theorem 7.2 (univalence)**.** *There is* $\mathrm{ua} \in \mathrm{Contr}(\mathsf{U}, \mathrm{Ua})$*, uniform as an equality of codes.*

The univalence operation is obtained from one universal datum.

Construction 7.3. The *universal univalence datum* $\mathcal{U}$ over GD is (SA, ext, extEquiv): the small copy of $\mathrm{Ext}_{\mathsf{U}}$ with $\mathrm{ext}_{\mathsf{U}}$ and its equivalence structure moved onto it, with codomain A_{U}. The code of SA along a map $s : \Delta \to \mathrm{GD}$ is given by a classifier element $\mathrm{smallCl}(s) \in \mathrm{Cl}(\Delta, s^*\mathrm{SA})$, the move along s of one universal element of $\mathrm{Cl}(\mathrm{GD}, \mathrm{SA})$, with

(C1) for $t : \Delta' \to \Delta$, the equality $\big((s^*\mathrm{SA})[t],\ \mathrm{smallCl}(s)[t]\big) = \big((st)^*\mathrm{SA},\ \mathrm{smallCl}(st)\big)$ in $\Sigma_A \mathrm{Cl}(\Delta', A)$, whose first component is Lemma 5.2; equivalently, $\mathrm{smallCl}(s)[t]$ is the transport of $\mathrm{smallCl}(st)$ along Lemma 5.2. It is the composite law of Lemma 3.9;

(C2) for $y \in \mathrm{GD}(W)$ full at the identity, $\big(\hat{y}^*\mathrm{SA},\ \mathrm{smallCl}(\hat{y})\big) = B_y(\mathrm{id})$ as elements of $\mathsf{U}(W)$.

When the classifier is Fib (as for posets), (C2) is proved once from two inputs of the instance: the fibrancy of El and Lemma 8.1. The type equality $\hat{y}^*\mathrm{SA} = \mathsf{El}(B_y)$ is not assumed but derived. On full points there is a cartesian map from $\mathsf{El}(B_y)$ into $\mathrm{Ext}_{\mathsf{U}}$ (the Yoneda map followed by the

inclusion of B on full points), and the fibres of $\hat{y}^*\mathrm{SA}$ are related to those of $\mathsf{El}(B_y)$ by a path whose transport, read in $\mathrm{Ext}_{\mathsf{U}}$, is that cartesian map. Since the cartesian map into $\mathrm{Ext}_{\mathsf{U}}$ is injective, this transport is natural, which gives the equality of types. The fibrations then agree by Lemma 3.10, Lemma 7.5(iii), Lemma 3.9 and Lemma 8.1.

Both steps of the proof below produce a full glue datum whose equivalence structure is a given structure moved along cartesian maps. Such a datum is determined by those maps.

Lemma 7.4 (uniqueness)**.** *Let $x \in \mathrm{GD}(X)$, and suppose its equivalence structure is the move of a structure ε along cartesian maps t_A, t_B that agree pointwise with the ones used to build a glue datum y from ε. If x and y have the same A, the same φ and the same codes B, then $x = y$.*

Proof. The maps α need no hypothesis: on each side both are read off the cartesian maps, by $t_A \circ \alpha = \mathrm{ext} \circ t_B$, so they agree once t_A and t_B agree pointwise. The two equivalence structures are moves of ε along pointwise equal maps, hence equal by Lemma 3.9. □

It is used with three facts about full points.

Lemma 7.5 (full points)**.** *(i) For every $d \in \mathrm{GD}(Z)$, e_d is e_{U} moved along the cartesian map of d's point into the universal datum.*

(ii) At a full point, the equivalence structure of $\mathrm{ext}_{\mathsf{U}}$ *moved along a cartesian map equals e_{U} moved along the composite of that map with the identification* $\mathrm{Ext} = B$.

(iii) On the full points, B maps into $\mathrm{Ext}(\alpha)$ *by a cartesian map, and* Ext*'s fibration moved along it is B's.*

Proof. (i) is the last clause of Lemma 6.4, which uses the coherence at every local point, not only at $o_x \cdot g$. (ii) and (iii) are the full-point clauses of Theorem 6.2, carried by Construction 3.7: a moved structure at ρ is $\tau^{-1}(k(s\rho, \varphi, \sigma u))$, and k at the full point $s\rho$ is given by those clauses; the remaining identification is uniqueness of moves (Lemma 3.9). For (iii): on a full line the composite of Ext is the end of B's filler (Theorem 6.2), which is B's composite by Lemma 4.3. □

Proof of Theorem 7.2. Let $F \in \mathsf{U}(X)$ and (φ, u) a partial element of Ua at F, $u_f = (B_f, \alpha_f, e_f)$. The proof has two steps.

(a) The datum of a partial element. There is a unique $d = (F, \varphi, B, \alpha, e) \in \mathrm{GD}(X)$ with $d \cdot f = u_f$ for f in φ. Its components B and α are read off the u_f at the identity. Its equivalence structure is e_{U} along $\ell_u : \varphi \to \mathrm{GD}$, $f \mapsto u_f$ (a map into the full points, the sieve φ read as a subpresheaf of yX), moved along the cartesian map of triples. So no new gluing is needed: the e_f were glued once, in e_{U} (Lemma 6.4), and the uniformity of u is what makes ℓ_u a map of presheaves.

(b) The extension. Let $\hat{d} : yX \to \mathrm{GD}$ and let $\mathrm{ua}(F, \varphi, u)$ be the full glue datum given by $\mathcal{U}$ along $\hat{d}$:

$$\mathrm{ua}(F, \varphi, u) = \hat{d}^*\mathcal{U} \text{ at the identity,}$$

with code $\mathrm{smallCl}(\hat{d})$ (at g, the code of $(d \cdot g)^*\mathrm{SA}$).

Steps (a) and (b) are two instances of one construction: a glue datum read through cartesian maps over a map of points into GD. Given B, a target triple with its equivalence structure, and cartesian maps on the A and B sides, it yields α, the naturality of α, and e. Its map is ext followed by the Yoneda isomorphism $\hat{d}^* A_{\mathsf{U}} \cong \mathsf{El}(F)$: the fibre of $\hat{d}^* A_{\mathsf{U}}$ at f is $\mathsf{El}(F \cdot (f \circ \mathrm{id}))$, equal to $\mathsf{El}(Ff)$ only up to $f \circ \mathrm{id} = f$, so this is a cartesian map whose fibre maps are transports; the equivalence structure is moved along it. *Uniformity*: $h^*(\varphi, u)$ gives dh, and $\widehat{dh} = \hat{d} \circ yh$; conclude by Lemma 5.2, Lemma 3.9 and (C1). *Sides*: for step (a), $d \cdot f = u_f$ on φ by Lemma 7.4 and

Lemma 7.5(i). For step (b), by uniformity it suffices to treat a side containing the identity; then $\varphi = \top$ and $d = u_{\mathrm{id}}$. The value of (b) has code B_{id} by (C2), map α_{id} (the Yoneda isomorphism is the identity at the identity), and its equivalence structure is e_{id} by Lemma 7.4 with Lemma 7.5(ii) and (i). Here the cartesian map of u_{id} into SA is the point map, then the identification $\mathrm{Ext} = B$ at full points, then the inverse of the comparison of SA with $\mathrm{Ext}_{\mathsf{U}}$; the inverse followed by the comparison is the identity by Lemma 3.9. The pointwise check on the B side uses only that the comparison is injective and that φ holds at full points. So the value is u_{id}. □

7.2 The universe is fibrant

Theorem 7.6. *For every context Γ, the constant type U over Γ has a fibration structure.*

Proof. Composition from e to e': let γ be a line of Γ, F a code at the start, and (φ, λ) a partial line of codes, with $\lambda_f(e) = Ff$. For each side, transport along λ_f from e' to e (Proposition 4.6, for El along λ_f) is an equivalence $\mathsf{El}(\lambda_f(e')) \to \mathsf{El}(Ff)$, natural in f. This is a partial element of Ua at F: concretely, the full glue datum read through cartesian maps over the map $f \mapsto \lambda_f$ into the lines of codes (the construction of step (a) of the proof of Theorem 7.2), so that its naturality is that of the construction. Its extension by ua has a code $Y \in \mathsf{U}(X)$, which is the composite. On a side, $Yf = \lambda_f(e')$ by the sides of ua; uniformity is that of ua and of transport. The equivalence structures must be natural in f strictly; this is where Proposition 4.6, and through it the fibrancy of path types over a line (Theorem 4.5), is used. □

Remark 7.7. This is cours6's route: the equivalence extension property gives univalence ($\Sigma(Y : \mathsf{U})\,\mathrm{Equiv}(Y, F)$ is contractible), and univalence gives $\mathrm{Comp}(1, \mathsf{U})$. No invariance under re-presentation of the cofibration is needed, because a partial element is on one sieve.

8 Instances

8.1 Posets

$\mathcal{C}$ is the category of finite non-empty posets and monotone maps (or the full subcategory on finite products of $[1]$), with $[1] = \{0 < 1\}$, connections max and min, and $\Phi(X)$ the decidable sieves on X. Finite unions are pointwise; $\forall$ along the interval is decided at the generic point $(\pi_1, \pi_2) : W \times [1] \to W \times [1]$, since $W \times [1]$ is an object.

The classifier is $\mathrm{Cl}(\Gamma, A) = \mathrm{Fib}(\Gamma, A)$, with the substitution of Section 4; it is coded because a composition is determined by its composites. Hypothesis 5.5 holds because $\mathbb{I}$ is tiny: $yX \times \mathbb{I} = y(X \times [1])$, so a line of U at X is a code over $y(X \times [1])$, with its own composition. The composition of El is glued by Lemma 6.4 from these: at a point x, the line's code at the generic point of $X \times [1]$, read in El through the Yoneda cartesian map; the coherence is that a code restricted along $g \times [1]$ is the code at xg, together with Lemma 8.1.

Lemma 8.1 (El along a Yoneda code)**.** *Let $F \in \mathsf{U}(X)$, with its fibration c_F (the classifier part of the code). Read El along the Yoneda map $\hat{F} : yX \to \mathsf{U}$; its fibre at f is $\mathsf{El}(F \cdot (f \circ \mathrm{id}))$, which is related to F by the Yoneda cartesian map (fibre maps: transport along $f \circ \mathrm{id} = f$). Then the fibration of El moved along this cartesian map is c_F.*

Proof. By Lemma 3.3 it suffices to compare composites. At a point f and a line over it, the glued composition of El is, by definition, the composition of the line's code at the generic point of $X \times [1]$; for a line of $\hat{F}$ this code is F restricted along $f \times [1]$, whose composition is c_F restricted, because

the classifier is moved by Construction 3.7. The two composites then agree through the cartesian map, by Lemma 3.9. □

The classifier is moved by Construction 3.7, so (C1) is Lemma 3.9. Condition (C2) holds by Theorem 6.2: at a full point every line over yX is full, and there Ext's composition is B's; this full-point composition clause is carried to SA by Lemma 7.5(iii). Everything is constructive.

8.2 Simplicial sets

$\mathcal{C} = \Delta$, $[1]$ the 1-simplex, connections max and min, $\Phi([n])$ the decidable sieves on $[n]$. Here $yX \times \mathbb{I}$ is not representable. $\forall$ along the interval is decided on the shuffles: the prism $[m] \times \Delta[1]$ is the union of the $m+1$ simplices $P_k : [m+1] \to [m] \times [1]$, and every simplex of the prism factors through one of them.

Definition 8.2 (horn filling)**.** For $0 \le k \le m+1$ let $\Lambda^k[m+1] \in \Phi([m+1])$ be the horn. A *horn filling operation* for A over Γ, $\mathrm{isFib}(\Gamma, A)$, assigns to each $\rho \in \Gamma([m+1])$ and each partial element of A at ρ on $\Lambda^k[m+1]$ an extension. No uniformity is required.

isFib is a structure, untruncated: an element is an operation. $\mathrm{Cl}(\Gamma, A) = \mathrm{isFib}(\Gamma, A)$ is a classifier: substitution moves a horn of $A[s]$ to a horn of A, fills it there and reads the filler back (Construction 3.7 for operations without uniformity).

Theorem 8.3.

1. *(constructive) A fibration structure gives a horn filling operation.*
2. *(with excluded middle) A horn filling operation gives a fibration structure.*

Proof. (1) A horn is a retract of the prism along a retraction line (Gabriel–Zisman): a family $r(f, i)$ with $r(f, e') = f$, $r(f, e)$ landing in one face of the horn, and r preserving the other faces. The composition along r extends the horn.

(2) First, *face-uniform* fillers, constructively: a filling problem at $[N]$ is solved by filling the shuffles $P_N, \dots, P_0$ (or $P_0, \dots, P_N$) one after the other, each by one horn filling; the result agrees with the sides and is uniform along face maps. Second, *normalised* fillers: a problem b is *degenerate* along s_i when $b = s_i^* d_i^* b$. This is a proposition, decided by excluded middle; the normalised filler of a degenerate problem is s_i^* of the filler of $d_i^* b$ for the first such i, and the face-uniform filler otherwise. By the Eilenberg–Zilber factorisation, by induction, normalised fillers are uniform along every map. They form a filling structure at each end, whose ends give the compositions. □

Excluded middle is used only in deciding degeneracy. The passage (2) is not natural in Γ (the first degeneracy is not preserved by substitutions identifying points), so the constructions never compose (2) with substitution: Hypothesis 5.5 is (2) applied once, to El, which has a horn filling operation by the classifier itself (a horn of El at a code (A, c) is a horn of A, filled by c).

The classifier of Ext**.** The code of SA is given by one horn filling operation on $\mathrm{Ext}_{\mathsf{U}}$, at level a^+, *by cases on fullness*: at a full point, the horn operation of B (read in B, since Ext is B there); otherwise, the horn operation of $\mathrm{Ext}_{\mathsf{U}}$'s composition given by Theorem 8.3(1). The case split is pointwise, and this is legitimate because a horn filling operation carries no uniformity: the universal element needs no naturality. Then $\mathrm{smallCl}(s)$ is this operation pulled back along s (Construction 3.7 for operations); (C1) is the composition law of moving operations. (C2) is proved by the same chain as for posets, with operations in place of fibrations: along the derived

equality $\hat{y}^*\mathrm{SA} = \mathsf{El}(B_y)$ (whose derivation does not use the classifier), Lemma 3.10 and the move laws for operations; at full points the universal operation is B's by the case split (the analogue of Lemma 7.5(iii)); and the analogue of Lemma 8.1: the operation of El moved along the Yoneda cartesian map of a code F is F's own. Its proof is definitional: at ρ, the operation of El at the code $F \cdot \rho$ is F's at $\rho \circ \mathrm{id}$, since the action on codes is substitution and the operation of a substitution is the pulled back one. The passage Theorem 8.3(2)$\circ$(1) is never used, and excluded middle reaches the final statements only through Hypothesis 5.5, i.e. through (2) applied once, to El. The univalence operation and Theorem 7.6 follow as in Section 7.

Remark 8.4 (choice). The usual Kan condition is a *property*: every horn merely has a filler. To obtain from it a horn filling *operation* requires the axiom of choice, and in univalent foundations choice yields only the mere existence of such an operation. The model uses the operation isFib as its definition and never passes through the property. How faithfully this represents the usual mathematics of Kan complexes, where filling is a property, is left for further analysis.

9 Identity types

The path type of Theorem 4.5 has an eliminator J defined by composition, but its computation rule holds only up to a path. We give two identity types with a strict computation rule: Swan's, for any base with one more property of Φ; and, on Δ with excluded middle, the path type itself.

Throughout, A is fibrant over Γ and $a, b \in \mathrm{Tm}(\Gamma, A)$. For $\rho \in \Gamma(X)$ an element $p \in \mathrm{Path}_A(a,b)(\rho)$ is a line $p(f,j) \in A(W, \rho f)$ over the constant line at ρ, with $p(e_0) = a\rho$ and $p(e_1) = b\rho$. The *constant path* at ρ is $(f,j) \mapsto a(\rho f)$; it is in $\mathrm{Path}_A(a,a)(\rho)$.

9.1 Swan's identity type

Hypothesis 9.1 (dependent conjunction)**.** Let $\psi \in \Phi(X)$ and let $\varphi_f \in \Phi(W)$, for $f : W \to X$ landing in ψ, be natural: $g^*\varphi_f = \varphi_{fg}$. Then the sieve

$$\psi \wedge \varphi = \{f \text{ in } \psi \mid \varphi_f = \top\}$$

is in $\Phi(X)$.

It is a sieve: if f is in ψ and $\varphi_f = \top$, then fg is in ψ and $\varphi_{fg} = g^*\varphi_f = \top$. It is decidable, since membership in ψ and $\varphi_f = \top$ are. In both instances $\Phi(X)$ is the set of all decidable sieves, so the hypothesis holds. Its two properties used below are: for f in ψ, $f^*(\psi \wedge \varphi) = \varphi_f$; and $h^*(\psi \wedge \varphi) = h^*\psi \wedge (g \mapsto \varphi_{hg})$.

Definition 9.2 (Id)**.** $\mathrm{Id}_A(a,b)(X,\rho)$ is the set of pairs (p, φ) with $p \in \mathrm{Path}_A(a,b)(\rho)$ and $\varphi \in \Phi(X)$ such that p is constant on φ: $p(f,j) = a(\rho f)$ for every f landing in φ and every j. Restriction along h is $(p,\varphi)h = (ph, h^*\varphi)$. Reflexivity is $\mathrm{refl}_a(\rho) = (\text{the constant path}, \top) \in \mathrm{Id}_A(a,a)(\rho)$, and $\pi : \mathrm{Id}_A(a,b) \to \mathrm{Path}_A(a,b)$ is the first projection.

This is a type: φ is coded by its characteristic map, and the condition is a proposition. On φ the two ends agree, $a(\rho f) = b(\rho f)$.

Proposition 9.3 (Id is fibrant)**.** $\mathrm{Id}_A(a,b)$ *has a fibration structure, natural in* Γ *in the sense of Theorem 4.5.*

Proof. Let γ be a line of Γ at X, (p_0, φ_0) an element over $\gamma(e)$ and (ψ, u) a partial line ending at it, with $u_f(j) = (p_{f,j}, \varphi_{f,j})$. Let p_1 be the composite of the path type (Theorem 4.5) for the partial line πu starting at p_0. The family $f \mapsto \varphi_{f,e'}$ (the sieve of the end of u_f) is natural on ψ; let $\varphi_1 = \psi \wedge \varphi_{-,e'}$. The composite is (p_1, φ_1).

It is an element: for f in φ_1, $p_1 f = p_{f,e'}$ by the sides of the path composition, and $p_{f,e'}$ is constant since $\varphi_{f,e'} = \top$. Sides: for f in ψ, $p_1 f = p_{f,e'}$ and $f^*\varphi_1 = \varphi_{f,e'}$. Uniformity: that of p_1, and $h^*\varphi_1$ is the dependent conjunction of the restricted data. Naturality: the composite is formed pointwise from the composition of the path type, which commutes with moves; the cartesian map of identity types is that of path types with the identity on φ. □

Theorem 9.4 (J, strictly)**.** *Let $\Gamma_a = \Gamma.A.\mathrm{Id}_A(a, q)$, with points (ρ, b, w), $w \in \mathrm{Id}_A(a\rho, b)$. Let C be a type over Γ_a with a composition $\mathrm{Comp}_{e_0 \to e_1}(C)$, and $d \in \mathrm{Tm}(\Gamma, C[a, \mathrm{refl}_a])$. There is $J(d) \in \mathrm{Tm}(\Gamma_a, C)$ with*

$$J(d)(\rho, a\rho, \mathrm{refl}_a\rho) = d(\rho).$$

Proof. Let $(\rho, b, (p, \varphi)) \in \Gamma_a(X)$ and let $c = c_0$ be the connection with $c(e_0, k) = e_0$ and $c(e_1, k) = k$. The line ℓ of Γ_a at X is

$$\ell(f, j) = \big(\rho f,\ p(f, j),\ (p^{f,j},\ f^*\varphi \cup (j = e_0))\big), \qquad p^{f,j}(g, k) = p(fg, c(jg, k)).$$

$p^{f,j}$ is a path from a to $p(f, j)$, constant on $f^*\varphi$ and on $(j = e_0)$, so $\ell(f, j)$ is a point; ℓ is natural. By the connection laws $\ell(e_0) = (\rho, a\rho, \mathrm{refl}_a\rho)$ and $\ell(e_1) = (\rho, b, (p, \varphi))$. For f in φ, ℓf is the constant line at $(\rho f, a\rho f, \mathrm{refl}_a\rho f)$. Define $J(d)(\rho, b, (p, \varphi))$ as the composite along ℓ from $d(\rho)$, with the side φ given by the constant lines at $d(\rho f)$. The side ends at $d(\rho)f$, as required.

$J(d)$ is natural because ℓ, φ and d restrict and the composition is uniform. At $(\rho, a\rho, \mathrm{refl}_a\rho)$, $\varphi = \top$ contains the identity, so by the sides of the composition the composite is the end of the side, $d(\rho)$. □

The computation rule is an equality of elements, and it is stable under substitution because both sides are natural. Only the composition from e_0 to e_1 of the motive is used.

Proposition 9.5 (Id and Path)**.** $\pi : \mathrm{Id}_A(a, b) \to \mathrm{Path}_A(a, b)$ *has an equivalence structure, natural in Γ.*

Proof. Let $q \in \mathrm{Path}_A(a, b)(\rho)$ and let (ψ, v) be a partial element of HFib_π at (ρ, q), $v_f = ((p_f, \varphi_f), \lambda_f)$ with λ_f a line of paths from p_f to qf. Fill the partial line λ by Fill_{e_1} of the path type, from q at e_1 (Proposition 4.2); let L be the filler and $p = L(e_0)$. Let $\varphi = \psi \wedge (f \mapsto \varphi_f)$. Then $((p, \varphi), L)$ is an element of the fibre: p is constant on φ because $pf = p_f$ for f in ψ. On ψ it is v_f, since $f^*\varphi = \varphi_f$ and $Lf = \lambda_f$; it is uniform because the filler and $\wedge$ are. □

So $\mathrm{Id}_A(a, b)$ and $\mathrm{Path}_A(a, b)$ are equivalent fibrant types, and Id has a strict J. The construction is generic: it uses Hypothesis 9.1, unions, the points of [1] and a connection, and holds for posets and for Δ.

9.2 The path type on Δ

A composition $\mathrm{Comp}_{e \to e'}(A)$ is *regular* if its composite along a constant line of Γ, from x, with a partial line constant in j, is x. With a regular composition of the motive, J for the path type, defined as in Theorem 9.4 without the cofibration ($\ell(f, j) = (\rho f, p(f, j), p^{f,j})$ and the empty side), computes strictly: at $(\rho, a\rho, \text{constant path})$ the line ℓ is constant.

Lemma 9.6. *A regular composition moves along a cartesian map to a regular composition.*

Proof. The maps σ send constant lines and constant partial lines to constant ones, and τ is injective. □

Regularity need not be preserved by the type formers. On Δ it does not have to be: a type of the model is a code, whose classifier part is a horn filling operation, and its composition is always that of El, moved along the code (Section 8.2). So it suffices to make the composition of El regular. Since a horn filling operation carries no uniformity, it can be changed pointwise.

Definition 9.7 (normal horn filling)**.** Horn data h on $\Lambda^k[n+1]$ at $\rho \in \Gamma([n+1])$ is *degenerate along* s_i, for $k \in \{i, i+1\}$, if $\rho = s_i^* \rho'$ and h is the restriction of $s_i^* y$ for some $y \in A(\rho')$. A horn filling operation is *normal* if it extends such h by $s_i^* y$.

The element y is the face of h at the vertex of $\{i, i+1\}$ other than k, so it is determined by h.

Lemma 9.8. *If h is degenerate along s_i and along s_l, then $s_i^* y = s_l^* y'$.*

Proof. If $i \neq l$, then $k \in \{i, i+1\} \cap \{l, l+1\}$; say $l = i+1 = k$. The faces i and $i+2$ are in the horn, so $y = d_i s_{i+1} y' = s_i d_i y'$ and $y' = d_{i+2} s_i y = s_i d_{i+1} y = s_i d_i y'$. Then $s_i^* y = s_i s_i d_i y'$ and $s_{i+1}^* y' = s_{i+1} s_i d_i y'$, which are equal. □

With excluded middle every horn filling operation has a *normalisation*: on horn data degenerate along some s_i (a proposition), the degenerate extension; otherwise the given one. By Lemma 9.8 the choice of i does not matter, and the normalisation is normal.

Proposition 9.9. *If the horn filling operation is normal, the fillers of Theorem 8.3(2) are regular: for a filling problem over a constant line at ρ, with sides constant in j and end x, the filler is the constant line at x. Hence both compositions are regular.*

Proof. By induction on the dimension N of the problem b. If b is degenerate along s_i, its filler is s_i^* of the filler of $d_i^* b$, a problem of the same kind, constant by induction. If a side contains the identity, the filler is that side, constant. Otherwise the prism is filled from its boundary and its lid at e: the faces are the fillers of the face problems, constant by induction, and the lid is x. So the data is the restriction of the constant prism $x \circ \mathrm{pr}$, and so is the horn data of the first shuffle P_k. Since $\mathrm{pr} \circ P_k = s_k$, this horn data is the restriction of $s_k^* x$ at $s_k^* \rho$; the missing face of the horn is d_k (the chain from e_0) or d_{k+1} (the chain from e_1), so the data is degenerate along s_k, and the normal operation fills P_k by $s_k^* x$, the constant prism along P_k. The same holds for each next shuffle. So the filler is the constant line, and its other end is x. □

Theorem 9.10. *On Δ, take as the fibration of El Theorem 8.3(2) applied to the normalisation of El's horn filling operation. Then the composition of every type $c^*\mathsf{El}$ is regular, and the path type has a J with strict computation rule for every motive of the model.*

Proof. The composition of $c^*\mathsf{El}$ is that of El moved along c; it is regular by Proposition 9.9 and Lemma 9.6. Nothing else in Section 8.2 changes: Hypothesis 5.5 only asks for a fibration of El, and condition (C2) concerns the horn filling operations, which are unchanged. □

Excluded middle is now used at two points, both in Section 8.2: deciding the degeneracy of filling problems, and of horn data. The normalisation is not natural in Γ (as for Theorem 8.3(2), the degeneracy of ρ is not preserved by substitutions), which is why it is applied once, to El. On posets the classifier is Fib itself: the composition of El at a code is the code's own, so regularity would have to be preserved by the type formers; there the strict identity type is Swan's.

A Correspondence with the formalisation

Files of `seg/core`; planned ones, from `seg/model`, in italics.

section	layer	files
1 Metatheory	L0	V, VCode, Encode, Presheaf; FunExtUA (function extensionality from univalence) and, in All, the instance ModelUA of the whole development under univalence alone
2 Base	L1	CBase (with $\top$), CSieve
3 Relative contractibility	L2	CPartial, CMove, CMoveLaws, CSub, COp, COpMove
4 Fibrations	L3	CLine, CConn, CFillProb, CFill, CCart, CFibSub, CPath, CEquiv; Path and transport (L7): CPathProb, CPathComp, CTr, CTrOm, CTrB, CTrP, CTrE; Σ, Π: CSig, CPiE, CPi; naturality: CFillMove, CSigNat, CPiCart, CPiNat; Path: CPathFib, CPathCart, CPathNat
5 Small types, universe	L4	CSmall, CUniv, CElC
6 Gluing, Theorem 7	L5	CExtL, CExt7Nat, CCof, CExt–CExt4, CExtEquiv–3, CExtComp1–4, CThm7, CExtFull, CLineSide, CMapLine, CHFibCart, CHFibLaws, CTrEq, CEnc, CGlueD, CGlueE, CGlueLaws, CGlueRC, CExtBig, CExtEq1–2, CExtFibU; *GExtCmp, GExtSmall*
7 Univalence, Comp$(1, \mathsf{U})$	L6, L7	L6: CHSub, CUa, CGlueCart, CGlueCartNat, CUaD, CUaB, CUaN, CUaBN, CGlueUniq, CUaFull1, CExtFullCart, CUaSA, CUaSB, CUaOp; L7: CFibU1, CFibU2, CFibU3
8.1 Posets	L8	Cat, CPosC, CPosBase, CPosInst, CPosEl1–5, CPosFinal; generic for Cl = Fib: CFibCl, CFibExt, CCartLaws, CSmallSub, CFillEnd, CExtFullFib, CTrFib, CYonCart, CSmallFull1–3, CFibFinal
8.2 Simplicial sets	L8	CDeltaBase, CSimp, CDeltaNat, CDeltaFact, CDeltaCl, CBoxHorn, CDeltaRetr, CDGlue–CDNormE, CDeltaHornBox, CDeltaHornToFib, CDElHorn, CDeltaElFib, CDGlueCl, CDSmallCl, CDSmallFull(2), CDOpFull, CDOpYon, CDeltaFinal; generic for operations: COpCl, COpTr, COpLaws, CHornRetr
9 Identity types	L3, L8	Swan's Id, generic: CId (type, refl, Hypothesis 9.1 as DConj), CJ (J from a contraction), CIdJ (Thm 9.4), CIdFib (Prop 9.3), CIdNat, CIdCart (its naturality), CIdEquiv (Prop 9.5), CIdEquivNat (its naturality); decidable sieves: CSieve.DConjS; instances: CPosId, CDeltaId. *Section 9.2 not yet formalised*